Additional file 5. Reported gonorrhoea prevalence and/or test positivity in men-who-have-sex-with-men, by WHO region, country, and anatomic site

WHO region	Country	Reference	Years reported	Gonorrhoea prevalence										
				Urethral			Rectal			Pharyngeal			Any/All	
				No. tested	Reported prevalence and/or test positivity	Standardized estimate	No. tested	Reported prevalence and/or test positivity	Standardized estimate	No. tested	Reported prevalence and/or test positivity	Standardized estimate	Total No. tested	Reported prevalence and/or test positivity
Africa	Botswana	Tafuma et al [1]	2012	450	1.4%	1.5%	450	1.8%	0.0%	..	..	..	450	2.9%
	Kenya	Sanders et al [2]	2011	244	1.6%	1.8%	244	5.7%	4.0%	..	..	..	..	..
		Sanders et al [3]	2005–2011	449	2.0%	2.2%	449	0.4%	0.6%	..	..	..	..	..
	Nigeria	Keshinro et al [4]	2013–2016	862	3.8%	4.4%	862	23.3%	22.9%	..	..	..	862	24.2%
	Senegal	Wade et al [5]	2007	500	2.6%	2.9%	..	..	..	..	..	..	..	..
	South Africa	Rebe et al [6]	2012	200	3.5%	4.0%	200	8.5%	7.0%	200	7.5%	5.9%	200	16.0%
	Tanzania	Ross et al [7]	2012–2013	219	1.0%	1.1%	219	12.3%	11.1%	..	..	..	..	..
	Uganda	Kim et al [8]	2008–2009	288	1.4%	1.5%	286	1.8%	0.0%	..	..	..	..	..
Americas	Brazil	Cunha et al [9]	2010–2012	273	0.0%	0.0%	279	2.5%	0.5%	..	..	..	..	..
(excluding	El Salvador	Creswell et al [10]	2008	648	1.2%	1.3%	506	3.6%	1.7%	..	..	..	..	..
high-	Jamaica	Figueroa et al [11]	2007–2008	201	3.5%	4.0%	..	..	..	..	..	..	..	..
income	Peru	Allan-Blitz et al [12][1]	2013–2014	..	..	..	387	8.8%	7.3%	387	6.5%	4.8%	..	..
North		Castillo et al [13]	2009–2012	..	..	..	510	8.6%	7.1%	510	5.4%	3.7%	..	..
America)		Kojima et al [14]	2013–2014	..	..	..	312	8.4%	6.9%	312	5.8%	4.1%	312	14.2%
		Leon et al [15]	2008–2009	..	..	..	701	9.6%	8.2%	712	6.5%	4.8%	..	..
		Perez-Brumer et al [16]	2007	560	2.1%	2.3%	..	..	..	..	..	..	..	..
Europe	Germany	Dudareva-Vizule et al [17]	2009–2010	685	1.9%	2.1%	2050	4.6%	2.8%	2197	5.5%	3.8%	..	..
		Marcus et al [18]	2011–2012	..	..	..	642	3.6%	1.7%	882	5.4%	3.7%	..	..
	Ireland	Igoe et al [19]	2013	..	..	..	..	..	..	..	..	..	3286	11.1%
	Italy	Foschi et al [20]	2017	..	..	..	165	27.3%	27.2%	84	19.0%	18.3%	..	..
	Netherlands	Heiligenberg et al [21]	2008–2009	673	4.8%	3.8%	673	5.1%	3.3%	673	4.2%	2.4%	673	10.7%
		Van Liere et al [22]	2010–2012	2436	1.5%	1.7%	2436	3.7%	1.9%	2436	3.4%	1.5%	2436	6.3%
		Van Liere et al [23]	2011–2012	9534	1.9%	0.9%	9534	4.2%	2.4%	532	90.6%	95.4%	..	..

	UK	Haidari et al [24][2]	2010–2012	..	..	..	..	..	..	..	..	..	..	13.0%
		Soni et al [25]	2008	438	0.2%	0.2%	412	3.2%	5.0%	..	..	..	438	5.0%
High-	Canada	Gratrix et al [26][3]	2012	972	2.4%	2.7%	972	5.9%	4.2%	..	..	..	972	6.5%
income		Remis et al [27]	2010–2012	147	0.0%	0.0%	..	..	..	..	..	..	..	..
North	USA	Anschuetz et al [28]	2014–2015	2756	2.4%	2.7%	1763	5.2%	3.5%	2505	5.6%	3.8%	..	..
America		Beymer et al [29]	2011–2013	..	..	..	..	..	..	..	..	..	6367	14.5%
		Crosby et al [30]	..	..	..	..	..	..	..	..	..	..	346	7.7%
		Freeman et al [31]	2009	..	..	..	..	..	..	480	6.7%	5.0%	..	..
		Hassan et al [32]	2013–2015	..	..	..	395	21.3%	20.7%	..	..	..	..	..
		Mayer et al [33]	2004–2010	889	0.1%	0.0%	..	..	..	..	..	..	..	..
		Montano et al [34]	2013–2014[4]	183	6.6%	5.5%	183	27.9%	27.8%	183	18.0%	17.3%	183	11.1%
		Mustanski et al [35]	2013–2015	998	0.9%	0.9%	995	5.0%	3.3%	..	..	..	..	..
		Park et al [36]	2010	..	..	..	..	..	..	12 457	5.8%	4.1%	..	..
		Patton et al [37]	2010–2012	18 460	11.1%	11.1%[5]	11 092	10.2%	8.9%	14 484	7.9%	6.4%	..	..
		Pinsky et al [38]	2007–2010	..	..	..	..	..	..	200	3.5%	1.6%	..	..
		Sexton et al [39]	2009–2011	..	..	..	374	8.0%	6.5%	374	9.3%	7.9%	..	..
		Taylor et al [40]	2011–2013	..	..	..	1591	19.6%	19.0%	..	..	..	..	..
South-East	India	Aggarwal et al [41]	2013–2014	52[6]	5.8%	6.2%	..	..	..	..	..	..	..	..
Asia	Indonesia	Hananta et al [42]	2014	275	17.8%	16.7%	..	..	..	..	..	..	..	...
		Morineau et al [43]	2007	749	1.9%	2.1%	738	18.6%	17.9%	..	..	..	..	..
	Thailand	Pattanasin et al [44]	2006–2010	1695	0.8%	0.9%	1552	5.9%	4.2%	1689	0.4%	0.0%	1695	6.4%
		Tongtoyai et al [45]	2006–2010	1743	1.8%	2.0%	1596	6.1%	4.5%	1740	0.5%	0.0%	..	..
Western	Australia	Cheung et al [46]	2007–2013	..	..	..	..	..	..	..	..	..	5256	27.8%
Pacific		Chow et al [47]	2015	..	..	..	..	..	..	823	10.1%	8.7%	..	..
		Chow et al [48]	2007–2013	25 766	2.3%	1.3%	22 746	2.9%	0.9%	25 233	1.7%	0.0%	..	..
		Nash et al [49]	2002–2012	37 553	1.7%	1.8%	..	..	..	..	..	..	..	..
		Ong et al [50]	2015–2016	5497	4.2%	4.7%	..	..	..	..	..	..	..	..
		Ryder et al [51]	2006–2008	5666	0.7%	0.0%	..	..	..	..	..	..	..	..
		Templeton et al [52]	2001–2004	..	..	..	..	..	..	1225	0.6%	0.0%	..	..

	Vodstrcil et al [53]	2002–2009	7611	4.2%	3.2%	5833	3.1%	3.1%[5]	6980	1.8%	0.0%	··	··
	Yang et al [54]	2015–2016	··	··	··	2392	10.5%	9.1%	2605	8.3%	6.7%	··	··
China[7]	Chen et al [55]	··	444	1.4%	1.4%[5]	··	··	··	··	··	··	··	··
	Fu et al [56]	2009	413	3.6%	4.1%	··	··	··	··	··	··	··	··
	Guo et al [57]	2007	··	··	··	··	··	··	··	··	··	139	15.8%
	Guo et al [58]	2009–2010	291	2.4%	2.7%	··	··	··	··	··	··	··	··
	Huan et al [59]	2010	··	··	··	328	3.7%	3.7%[5]	··	··	··	··	··
	Liu et al [60]	2009	1008	3.0%	3.0%[5]	··	··	··	··	··	··	··	··
	Yang et al [61]	2015–2017	463	5.2%	6.0%	463	6.1%	4.4%	463	3.9%	2.0%	463	12.5%
	Zhang et al [62]	2014–2015	296	1.4%	1.5%	296	6.8%	5.2%	296	8.1%	6.6%	296	13.2%
Korea (Rep. of)	Jung et al [63]	2008	106	0.0%	0.0%	··	··	··	··	··	··	··	··
Vietnam	Pham et al [64]	2009	381	1.8%	2.0%	··	··	··	··	··	··	··	··

If the standardized estimate was a negative number, the standardized prevalence was reported at 1 case divided by 100 times the sample size [65].

[1]85 (22%) study participants were transgender. The results for MSM and transgender were not reported separately.

[2]Proportion of MSM tested during routine testing at Department of Genitourinary Medicine. Denominator was not provided in this letter to the editor.

[3]Testing prevalence per screening visit is reported. Number of study participants not reported.

[4]The proportion of MSM with gonorrhoea over a 12-month period prior to starting pre-exposure prophylaxis is reported.

[5]Unable to apply laboratory standardization due to lack of information on laboratory testing.

[6]Given the rarity of data from India, this study was included despite a sample size of <100.

[7]Two additional studies were reported in a systematic review by Chow et al [66], for which we were unable to retrieve the original reports of the data. According to Chow et al, prevalence of gonorrhoea in MSM was reported to be 1.4% (1/70) in Yunnan Province in 2009 and 1.0% (1/105) in 2010 [67]; in 2008 in Nanning, prevalence of urethral infection in MSM was 0.9% (1/106) and of rectal infection 0% (0/130) [68].

··=not reported. MSM=men-who-have-sex-with-men. No=number. Rep.=Republic. UK=United Kingdom. USA= United States of America. WHO=World Health Organization.

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